

NOTE ON REGULARITY THEORY FOR NONLINEAR ELLIPTIC EQUATIONS

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ABSTRACT. In this note, we present several seminal developments in the *regularity* theory of nonlinear (uniformly) elliptic equations, including the De Giorgi-Nash-Moser theory concerning the Hilbert 19th problem and variational equations, as well as the Krylov-Safonov and Evans-Safonov theories for fully nonlinear equations.

This note benefited greatly from seminars discussions with my friends. I acknowledge them for their valuable perspectives.

1. INTRODUCTION

This note provide a unified exposition of three landmark contributions to the *regularity* theory of second-order elliptic equations: De Giorgi-Nash-Moser theory of divergence-type equations (1957), the Krylov-Safonov Harnack inequality for non-divergence equations (1979), and the Evans-Krylov interior $C^{2,\alpha}$ estimates for concave fully nonlinear equations (1982). All of them make a jump of invariance classes instead of small perturbation.

The general setting is the equation

$$F(D^2u, Du, u, x) = 0,$$

which, under the Schauder theory ([Wan06]) and suitable conditions on the lower-order terms, can often be reduced to the fully nonlinear model

$$F(D^2u) = 0.$$

We shall always assume that F is *uniformly* elliptic and smooth; the objective is to deduce higher regularity of the solution u .

Historically, the problem originates with Hilbert's nineteenth problem (1900), which asked whether minimizers of a uniformly convex energy

$$\mathcal{E}[u] = \int F(\nabla u) dx$$

must be smooth. The Euler–Lagrange equation of such a variational problem is of divergence type,

$$D_j (F_{ij}(\nabla u) D_i u_e) = 0,$$

and De Giorgi's breakthrough [DG57] consisted in combining an energy estimate with the Sobolev inequality in an iterative scheme that yields $C^{1,\alpha}$ regularity, thereby initiating the *bootstrap*.

For equations that are not in divergence form, the energy estimate is unavailable. Instead, the theory relies on the *maximum principle* and on measure estimates of the contact set, initiated by the works of Calderón–Zygmund (see [MS13]) as well as Alexandroff [Ale58], Bakelman [Bak83] and Pucci (see [PS07]). Building on these ideas, Krylov and Safonov [KS79, KS81] established a *Harnack* inequality for solutions of

$$a_{ij}(x) D_{ij} u = 0, \quad \lambda \delta_{ij} \leq a_{ij} \leq \Lambda \delta_{ij},$$

with merely measurable coefficients, which immediately implies $C^{1,\alpha}$ regularity.

When the equation is both uniformly elliptic and concave (or convex), Evans [Eva80, Eva82] and Krylov [Kry83, Kry84] established interior $C^{2,\alpha}$ regularity. Their proof hinges on the observation that second derivatives, say v , behave as "almost supersolutions." This property allows one to apply the Krylov–Safonov L^ϵ -estimate to deduce oscillation decay for v , which in turn yields the $C^{2,\alpha}$ estimate. Bootstrap arguments then imply that the solution is in fact smooth.

These three theories—De Giorgi–Nash–Moser, Krylov–Safonov, and Evans–Krylov—form the core of modern elliptic regularity theory and have inspired countless extensions to degenerate, nonlocal, and geometric settings.

Structure of the notes.

1. De Giorgi–Nash–Moser theory for divergence-type equations.
2. Krylov–Safonov theory for non-divergence equations.
3. Evans–Krylov theory for concave fully nonlinear equations.

The presentation follows the spirit of several well-known lecture notes, including those of Caffarelli [CV10], Vasseur [Vas16], professor and Yu Yuan’s lecture notes, as well as the classical monograph [CC95]. The material was also shaped by a series of inspiring lectures given by Professor Jiakun Liu at the 2024 summer school of BICMR.

2. HILBERT 19TH PROBLEM AND DE GIORGI–NASH–MOSER THEORY

2.1. Introduction to Hilbert 19th problem and calculus of variations. We begin with Hilbert’s 19th problem, proposed at the 1900 ICM [Hil00]. Consider the functional

$$\mathcal{E}[u] := \int_{\Omega} F(\nabla u) dx, \quad u \in H^1(\Omega)$$

where $F : \mathbb{R}^n \rightarrow \mathbb{R}$ is smooth and *uniformly convex*, and $\Omega \subset \mathbb{R}^n$. The question is: Are local minimizers of $\mathcal{E}$ necessarily smooth?¹ This conjecture was motivated by the Dirichlet energy $E[u] := \int_{\Omega} |\nabla u|^2$ (with $F(p) = |p|^2$), whose Euler-Lagrange equation $\Delta u = 0$ yields harmonic functions. By Weyl’s lemma [HL11], any H^1 solution is smooth, leading to the conjecture for general $F(\nabla u)$, a more challenging case.

Recall that $u \in H^1$ is a *local minimizer* of $\mathcal{E}$ if,

$$\mathcal{E}[u] \leq \mathcal{E}[u + \varphi], \quad \forall \varphi \in C_c^\infty(\Omega).$$

¹Hilbert’s original statement: “*there exist partial differential equations whose integrals are all of necessity analytic functions of the independent variables, that is, in short, equations susceptible of none but analytic solutions.*”

The functional $\mathcal{E}$ and F are uniformly convex if,

$$(2.1) \quad 0 < \lambda I \leq D^2 F(p) \leq \Lambda I, \quad \forall p \in \mathbb{R}^n, \text{ for some universal positive } \lambda, \Lambda.$$

Since u minimizes $\mathcal{E}$, consider a perturbation $u + \epsilon\varphi$ and differentiate:

$$0 = \frac{d}{d\epsilon} \Big|_{\epsilon=0} \int_{\Omega} F(\nabla u + \epsilon \nabla \varphi) = \int_{\Omega} D_j F(\nabla u) D_j \varphi, \quad \forall \varphi \in C_c^\infty(\Omega).$$

Thus, u weakly satisfies the Euler-Lagrange equation:

$$(2.2) \quad D_j(D_j F(\nabla u)) = 0, \quad \text{in } \Omega,$$

which is of divergence type. Differentiating (2.2) in direction $e \in \mathbb{S}^{n-1}$ gives the linearized equation:

$$(2.3) \quad D_j(F_{ij}(\nabla u) D_i u_e) = 0,$$

where $F_{ij} = D_{ij} F$, $u_e = D_e u$. By (2.1), both equations are *uniformly elliptic* (throughout the whole note).

To establish smoothness, it suffices to show $u \in C^{1,\alpha}$, initiating a *bootstrap* argument. If $u \in C^{1,\alpha}$, then $F_{ij}(\nabla u) \in C^{0,\alpha}$, and by *Schauder* theory², $u \in C^{2,\alpha}$. Iterating yields smoothness. Let $v = u_e$ and $a_{ij}(x) = F_{ij}(\nabla u)$. Then (2.3) becomes

$$(2.4) \quad D_j(a_{ij}(x) D_i v) = 0, \quad v \in H^1(\Omega),$$

with measurable coefficients a_{ij} . Note that $u \in C^{1,\alpha}$ iff $v \in C^\alpha$. Thus, Hilbert's problem reduces to whether solutions of (2.4) are in C^α .

Early contributions include [Ber04, Sch34, Mor38]. In 1957, De Giorgi [DG57] solved the problem via an iteration method, which will be presented in this paper. Nash [Nas57, Nas58] treated the parabolic case, and Moser [Mos60] provided an alternative

²We refer readers to [HL11, GT77] for more discussions on classical theories.

method to lift integrability to L^∞ . These form the **De Giorgi-Nash-Moser theory** (DNM theory) for equations like (2.4).

We demonstrate the main theorem of this section.

Theorem 1 (De Giorgi, 1957 [DG57]). Let $v \in H^1(B_1)$ be a solution to (2.4), i.e.,

$$D_j(a_{ij}(x)D_i v) = 0,$$

where $a_{ij}(x)$ are uniformly elliptic and measurable. Then

$$\|v\|_{C^\alpha(B_{1/2})} \leq C\|v\|_{L^2(B_1)},$$

where C depends on n, λ, Λ .

The proof is divided into two parts, say the *iteration* and *oscillation* lemma, and are demonstrated in the following two subsections, respectively, based on [CV10, Vas16].

To set the stage for the regularity theory, we first establish the existence and uniqueness of local minimizers [FRRO23].

Proposition 1 (Existence and uniqueness of local minimizer). Let Ω be a bounded Lipschitz domain and suppose

$$(2.5) \quad \{w \in H^1(\Omega) : w = g \text{ on } \partial\Omega\} \neq \emptyset.$$

Then there exists a unique local minimizer $u \in H^1(\Omega)$ with $u|_{\partial\Omega} = g$ for $\mathcal{E}$. Moreover, u weakly satisfies (2.2).

Proof. By (2.1), without loss of generality, we may assume F attains its minimum at 0 with $F(0) = 0$, so $DF(0) = 0$ and

$$\lambda|p|^2 \leq D^2F(p) \leq \Lambda|p|^2, \quad \forall p \in \mathbb{R}^n.$$

For existence, let

$$\mathcal{E}_0 := \inf\{\mathcal{E}[w] : w|_{\partial\Omega} = g, w \in H^1(\Omega)\},$$

which is finite since $\mathcal{E}_0 \leq \Lambda\|w\|_{\dot{H}^1(\Omega)} < \infty$. Take a minimizing sequence $\{w_k\} \subset H^1$ with $\mathcal{E}[w_k] \rightarrow \mathcal{E}_0$. Then $\|\nabla w_k\|_{L^2(\Omega)} \leq \lambda^{-1}\mathcal{E}[w_k] < \infty$, and by Poincaré's inequality, $\{w_k\}$ is bounded in $H^1(\Omega)$. The Rellich-Kondrachov theorem [Eva82] and Banach-Alaoglu theorem [BB11] yield a subsequence $\{w_{k_j}\}$ converging strongly in $L^2(\Omega)$ and weakly in $H^1(\Omega)$ to some $w \in H^1(\Omega)$.

To show w is the minimizer, we prove weak lower semi-continuity of $\mathcal{E}$:

$$\mathcal{E}[w] \leq \liminf_{j \rightarrow \infty} \mathcal{E}[w_{k_j}] = \mathcal{E}_0.$$

Define $\mathcal{A}(t) := \{v \in H^1(\Omega) : \mathcal{E}[v] \leq t\}$, which is convex. If $v_k \rightarrow v$ strongly in H^1 with $\{v_k\} \subset \mathcal{A}(t)$, then Fatou's lemma gives

$$\mathcal{E}[v] = \int_{\Omega} F(\nabla v) \leq \liminf_{k \rightarrow \infty} \int_{\Omega} F(\nabla v_k) \leq t,$$

so $\mathcal{A}(t)$ is strongly closed. By Mazur's theorem [BB11], $\mathcal{A}(t)$ is weakly closed. Now let $t^* := \liminf_{k \rightarrow \infty} \mathcal{E}[w_k]$. For any $\epsilon > 0$, there exists a subsequence $\{w_{k_j, \epsilon}\}$ with $\mathcal{E}[w_{k_j, \epsilon}] \leq t^* + \epsilon$. Since $\mathcal{A}(t^* + \epsilon)$ is weakly closed, $\mathcal{E}[w] \leq t^* + \epsilon$. Letting $\epsilon \rightarrow 0^+$ yields the result.

For uniqueness, suppose $u, v \in H^1(\Omega)$ are two minimizers with $u|_{\partial\Omega} = v|_{\partial\Omega} = g$. If $u \neq v$, there exists $D \subset \Omega$ with $|D| > 0$ such that $\nabla u \neq \nabla v$ on D . By uniform convexity,

$$\frac{F(\nabla u) + F(\nabla v)}{2} > F\left(\frac{\nabla u + \nabla v}{2}\right) \quad \text{on } D.$$

Integrating over Ω gives

$$\mathcal{E}_0 = \int_{\Omega} \frac{F(\nabla u) + F(\nabla v)}{2} > \int_{\Omega} F\left(\frac{\nabla u + \nabla v}{2}\right) \geq \mathcal{E}_0,$$

which leads to the contradiction. $\square$

2.2. DNM theory I: De Giorgi-Moser iteration. We now begin the first part of the proof, which establishes an $L^2 \rightarrow L^\infty$ estimate. While such an estimate is generally nontrivial, it is expected for solutions of elliptic equations.

Lemma 1. Suppose that

$$\|v^+\|_{L^2(B_1)} < \delta_0 \ll 1,$$

where δ_0 is sufficiently small. Then

$$\|v^+\|_{L^\infty(B_1)} \leq 1.$$

De Giorgi's iteration combines two key ingredients: the Sobolev inequality and an *energy* inequality, which together yield an iterative scheme by the L^2 and L^∞ norms. We present the statements of these lemmas following Caffarelli's Lecture note.

The first lemma applies to general functions:

sublemma 1 (Sobolev inequality). Let $w \in H^1(\mathbb{R}^n)$ have compact support. Then

$$\|w\|_{L^p} \lesssim \|\nabla w\|_{L^2},$$

where $2 < p \leq \frac{2n}{n-2}$.

Remark 1. The lower bound $p > 2$ is crucial for the iteration.

Proof. We first establish the representation

$$(2.6) \quad |w(x_0)| \leq \left| \int_{\mathbb{R}^n} \frac{\nabla w(y) \cdot (x_0 - y)}{|x_0 - y|^n} dy \right|.$$

Without loss of generality, set $x_0 = 0$. Then

$$\begin{aligned} w(0) &= -\frac{d}{dt} \frac{1}{|\partial B_1|} \int_{\mathbb{S}^{n-1}} \int_0^\infty w(t\nu) dt d\nu \\ &= -\frac{1}{|\partial B_1|} \int_{\mathbb{S}^{n-1}} \int_0^\infty \nabla w(t\nu) \cdot \nu dt d\nu \\ &= \int_{\mathbb{R}^n} \frac{\nabla w(y) \cdot (0 - y)}{|0 - y|^n} dy. \end{aligned}$$

Define $G(x_0, y) = \frac{x_0 - y}{|x_0 - y|^n}$. Taking the L^p norm of (2.6) and applying Hölder inequality yields

$$\begin{aligned} \int |w(x_0)|^p dx_0 &\leq \int \left| \int_{\mathbb{R}^n} \nabla w(y) \cdot G(x_0, y) dy \right|^p dx_0 \\ &\leq \|\nabla w\|_{L^2}^p \int \|G(x_0, y)\|_{L_y^2}^p dx_0. \end{aligned}$$

Since $\|G(x_0, y)\|_{L_y^2}$ is p -integrable for $2 < p \leq \frac{2n}{n-2}$, the result follows. $\square$

Remark 2. Replacing the exponent 2 by a general $q < n$ in the proof yields the Sobolev inequality for $p = \frac{nq}{n-q}$.

The second lemma concerns subsolutions:

sublemma 2 (Energy inequality). Let nonnegative $v \in H^1$ be a *subsolution* to (2.4), i.e.,

$$D_j(a_{ij}(x)D_i v) \geq 0, \quad \text{in } B_1$$

Then for any cut-off function $\varphi \in C_c^\infty(B_1)$, we have

$$\int |\nabla(\varphi v)|^2 \lesssim C \sup |\nabla \varphi|^2 \int (v)^2.$$

Proof. Multiply the subsolution inequality by $\varphi^2 v \geq 0$ and integrate by parts:

$$\int a_{ij}(2\varphi v D_i \varphi D_j v + \varphi^2 D_i v D_j v) \leq 0.$$

Hence

$$\begin{aligned} \int a_{ij} D_i(\varphi v) D_j(\varphi v) &= \int a_{ij} [(2\varphi v D_i \varphi D_j v + \varphi^2 D_i v D_j v) + D_i \varphi D_j \varphi v^2] \\ &\leq \int a_{ij} D_i \varphi D_j \varphi v^2. \end{aligned}$$

Using the uniform ellipticity $0 < \lambda \delta_{ij} \leq a_{ij} \leq \Lambda \delta_{ij}$, we obtain

$$\lambda \int |\nabla(\varphi v)|^2 \leq \Lambda \sup |\nabla \varphi|^2 \int v^2,$$

which completes the proof. $\square$

Remark 3. The domain of integration on the right-hand side can be restricted to $B_1 \cap \text{supp } \varphi$.

We now prove Lemma 1 using an iterative method.

Proof of Lemma 1. Define cutoff functions $\varphi_k \in C_c^\infty(B_1)$ satisfying

$$\varphi_k := \begin{cases} 1, & \text{in } B_{1/2+2^{-k}} \\ 0, & \text{outside } B_{1/2+2^{-k+1}} \end{cases}, \quad |\nabla \varphi_k| \lesssim 2^k, \quad k = 1, 2, \dots.$$

Define the *truncated* functions $v_k := (v - (1 - 2^{-k}))^+$ for $k = 1, 2, \dots$. We shall see that $\varphi_k v_k \rightarrow 0$ as $k \rightarrow \infty$.

By the Sobolev inequality,

$$\left(\int_{B_1} (\varphi_k v_k)^p \right)^{2/p} \lesssim \int_{B_1} |\nabla(\varphi_k v_k)|^2.$$

Since $p > 2$, Hölder inequality yields

$$\int_{B_1} (\varphi_k v_k)^2 \lesssim \left(\int_{B_1} (\varphi_k v_k)^p \right)^{2/p} \cdot |\{\varphi_k v_k > 0\}|^\epsilon,$$

where $\epsilon = 1 - \frac{2}{p} > 0$. Letting $A_k := \int (\varphi_k v_k)^2$, we obtain

$$A_k \lesssim |\{\varphi_k v_k > 0\}|^\epsilon \cdot \int_{B_1} |\nabla(\varphi_k v_k)|^2 =: |\{\varphi_k v_k > 0\}|^\epsilon \cdot E_k.$$

We now estimate E_k using the energy inequality. Note that $\{\varphi_k > 0\} \subset \{\varphi_{k-1} \equiv 1\} \cap \{v_{k-1} > v_k\}$, hence

$$E_k \lesssim \sup |\nabla \varphi_k|^2 \int_{B_1 \cap \text{supp } \varphi_k} (v_k)^2 \leq 2^{2k} \int (\varphi_{k-1} v_{k-1})^2 = 2^{2k} A_{k-1}.$$

Next, observe that $\{\varphi_k v_k > 0\} \subset \{\varphi_{k-1} v_{k-1} > 2^{-k}\}$. Chebyshev inequality gives

$$|\{\varphi_k v_k > 0\}| \leq |\{\varphi_{k-1} v_{k-1} > 2^{-k}\}| \leq 2^{2k} \int (\varphi_{k-1} v_{k-1})^2 = 2^{2k} A_{k-1}.$$

Combining these estimates yields

$$A_k \lesssim 2^{2k(1+\epsilon)} A_{k-1}^{1+\epsilon}, \quad \epsilon > 0.$$

If $A_1 \leq \delta_0$ is sufficiently small, then $A_k \rightarrow 0$ as $k \rightarrow \infty$, completing the proof. $\square$

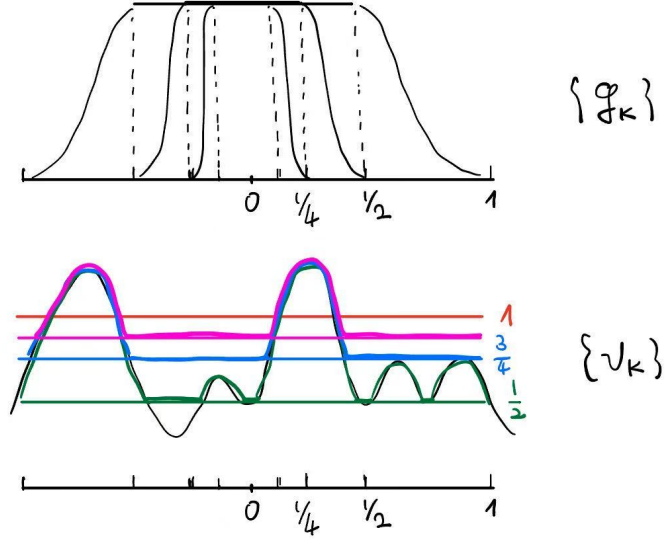


FIGURE 1. cutoff functions

Corollary 1.

$$\|v\|_{L^\infty(B_{1/2})} \lesssim \|v\|_{L^2(B_1)}$$

Remark 4. Note that we only use the subsolution v .

2.3. DNM theory II: Oscillation lemma. Define the oscillation of a function v over Ω by

$$\text{osc}_\Omega v := \sup_\Omega v - \inf_\Omega v.$$

Lemma 2 (Oscillation decay). Let v be a solution to (2.4) in B_1 . Then

$$\text{osc}_{B_{1/2}} v \leq \sigma \text{osc}_{B_1} v,$$

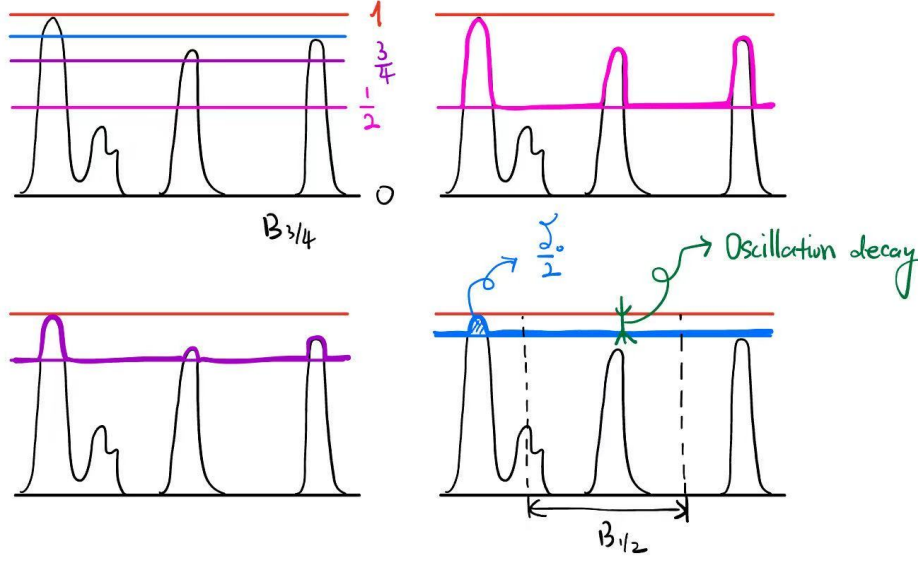
where $\sigma \in (0, 1)$ is a universal constant.

Remark 5. This lemma implies $v \in C^\alpha(B_{1/2})$, completing the proof of Theorem 1.

Lemma 1 ensures that when $\|v\|_{L^2(B_1)}$ is sufficiently small, we have $\text{osc}_{B_1} v \leq 2$; in particular, $\text{osc}_{B_{3/4}} v \leq 2$. Now suppose $v^+ \equiv 0$ in $B_{3/4}$ except on a set of measure at most $\delta_0/2$, which implies $\|v^+\|_{L^2(B_{3/4})} \leq \delta_0/2$. Then Lemma 1 yields $\text{osc}_{B_{1/2}} v \leq 1$. Note that either v^+ or v^- must vanish on at least half of $B_{3/4}$, with loss of generality,

$$\frac{|\{v^+ \equiv 0\} \cap B_{3/4}|}{|B_{3/4}|} \geq 1/2.$$

Thus, it remains to establish the vanishing of v^+ on a sufficiently large subset. The idea is to iteratively *truncate* and *renormalize* v^+ so that the function vanishes on a progressively larger set, ultimately reaching the desired large measure.

FIGURE 2. truncate v^+

First, we shall show that for any H^1 function, the set where it remains bounded away from both its supremum and infimum has a quantitatively positive measure, enabling such truncation and renormalization.

sublemma 3 (De Giorgi isoperimetric inequality). Let $w \in H^1(B_1)$ satisfy

$$\int_{B_1} |\nabla w|^2 \leq 1, \quad 0 \leq w \leq 1, \quad w \text{ touches } 1 \text{ at some points.}$$

Define the sets

$$A := \{w = 0\} \cap B_1, \quad C := \{w = 1\} \cap B_1, \quad D := \{0 < w < 1\} \cap B_1.$$

Then

$$|D| \gtrsim |C|^{2-\frac{2}{n}}.$$

Proof. Fix $x_0 \in A$ and let cone A be the cone over A with vertex x_0 in B_1 , and let $S(A)$ denote the surface area of cone $A \cap \partial B_1$. See Figure 3. Then

$$1 = w(x_0) = \frac{1}{S(A)} \int_{\text{cone } A} \frac{\nabla w(y) \cdot (x_0 - y)}{|x_0 - y|^n} dy,$$

or

$$S(A) \leq \int_{\text{cone } A} \frac{|\nabla w(y)|}{|x_0 - y|^{n-1}} dy.$$

By the *isoperimetric* inequality, $|A|^{(n-1)/n} \lesssim S(A)$. Since $|\nabla w|$ vanishes on $A \cup C$, we have

$$|A|^{(n-1)/n} \lesssim \int_D \frac{|\nabla w(y)|}{|x_0 - y|^{n-1}} dy.$$

Integrating over $x_0 \in C$ and applying Fubini theorem yields

$$|A|^{(n-1)/n} \cdot |C| \lesssim \int_C \int_D \frac{|\nabla w(y)|}{|x_0 - y|^{n-1}} dy dx_0 = \int_D |\nabla w(y)| \left(\int_C \frac{1}{|x_0 - y|^{n-1}} dx_0 \right) dy.$$

Note that

$$\int_C \frac{1}{|x_0 - y|^{n-1}} dx_0 \leq \frac{1}{|\partial B_1|} \int_{\mathbb{S}^{n-1}} \int_0^{|C|^{1/n}} \frac{1}{r^{n-1}} r^{n-1} dr d\theta = |C|^{1/n},$$

and by Hölder inequality,

$$\int_D |\nabla w| dy \leq \|\nabla w\|_{L^2(B_1)} \cdot |D|^{1/2} \leq |D|^{1/2}.$$

Thus,

$$|A|^{(n-1)/n} \cdot |C| \lesssim |D|^{1/2} \cdot |C|^{1/n}.$$

Noticing that $|A| \gtrsim |B_1|/2$, we obtain

$$|D| \gtrsim |C|^{2-\frac{2}{n}},$$

completing the proof. □

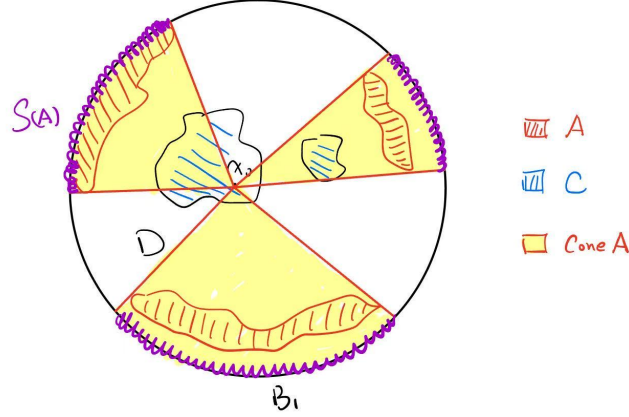


FIGURE 3. measure estimate

Remark 6. The proof extends to $W^{1,p}$ functions for $1 < p < \infty$.

Finally, we prove Lemma 2.

Proof of Lemma 2. Assume without loss of generality that $\sup_{B_1} v^+ \leq 1$. Define the rescaled functions

$$v_k := \frac{v^+ - (1 - 2^{-k})}{2^{-k}}, \quad k = 0, 1, 2, \dots$$

and the corresponding sets

$$A_k := \{v_k = 0\} \cap B_{3/4}, \quad C_k := \{v_k = 1\} \cap B_{3/4}, \quad D_k := \{0 < v_k < 1\} \cap B_{3/4}.$$

We prove that for some k_0 , the function v_{k_0} vanishes in $B_{3/4}$ except on a set of measure at most $\delta_0/2$. Suppose otherwise, and assume $|C_1| \geq \delta > 0$. Then by iteration,

$$|A_1| = |A_0| + |D_1| \gtrsim \frac{1}{2}|B_{3/8}| + \delta^{2-\frac{2}{n}},$$

$$|A_k| \gtrsim \frac{1}{2}|B_{3/8}| + k\delta^{2-\frac{2}{n}}, \quad k = 2, 3, \dots,$$

which yields a contradiction for sufficiently large k . □

Finally, we establish a *Liouville-type* result as a direct consequence.

Corollary 2. Let v be an entire solution of (2.4) in $\mathbb{R}^n$, with

$$\|v\|_{L^\infty} \leq C.$$

Then v is constant.

Sketch of proof. Iterating the oscillation estimate yields

$$\text{osc}_{B_1} v \leq \sigma^k \text{osc}_{B_{2^k}} v \rightarrow 0 \quad \text{as } k \rightarrow \infty,$$

completing the proof. □

Remark 7. Thus, if the *global* minimizer u satisfies $\|\nabla u\|_{L^\infty} \leq C$, then u is a plane.

2.4. Density estimate for minimal surfaces. De Giorgi's iteration scheme proves powerful in establishing decay estimates when two quantities of different homogeneities compete, such as integrability versus differentiability in function theory, or *volume* versus *perimeter* in minimal surfaces and modern geometric measure theory.

As an example, we present an iteration argument in the geometric setting of minimal surfaces. First, recall a well-known density estimate for minimal surfaces [CM11, GW84].

Theorem 2 (Density estimate). Let $0 \in \partial E$ be a minimal hypersurface in $\mathbb{R}^{n+1}$. Then there exists a universal constant $\delta_0 > 0$ such that

$$|B_1 \cap E| \geq \delta_0.$$

Proof. We argue by contradiction. Suppose $|B_1 \cap E| < \delta_0$. We show that $B_{1/2} \cap E = \emptyset$.

First, we recall two key tools:

A (Isoperimetric inequality) Since ∂E is area-minimizing in B_1 , the isoperimetric inequality gives

$$(2.7) \quad |B_r \cap E| \lesssim P_{B_r}(E)^{\frac{n}{n-1}} \leq |2S|^{\frac{n}{n-1}},$$

where $P_{B_r}(E)$ denotes the perimeter of E in B_r , and $S := S_r$ is the intersection of E with ∂B_r .

B (**Excess estimate**) Let S_ρ denote the ρ -slice of $(B_R \setminus B_r) \cap E$. Then

$$|(B_R \setminus B_r) \cap E| = \int_r^R |S_\rho| d\rho \geq \min_\rho |S_\rho| \cdot (R - r),$$

so that

$$(2.8) \quad \min_\rho |S_\rho| \leq \frac{1}{R - r} |(B_R \setminus B_r) \cap E|.$$

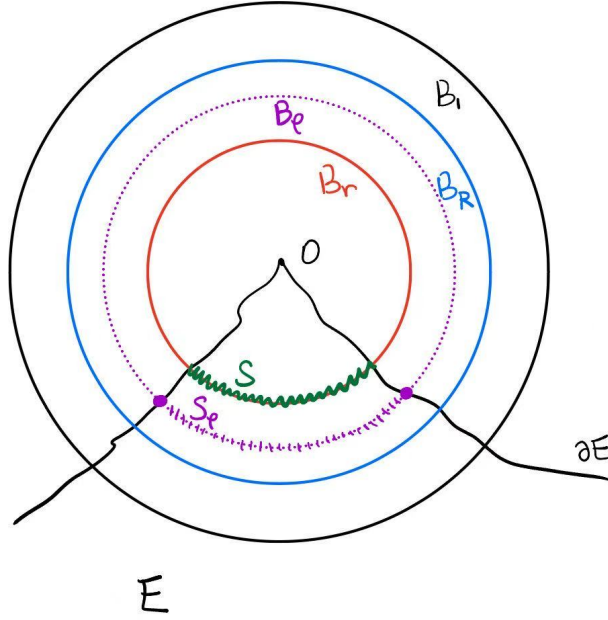


FIGURE 4. Minimal surface.

We now show that $|B_{1/2+2^{-k}} \cap E| \rightarrow 0$ as $k \rightarrow \infty$. Let S_k be the area-minimizing ρ -slice between $B_{1/2+2^{-k-1}}$ and $B_{1/2+2^{-k}}$, and let $\mathcal{C}_k$ be the cone with vertex 0 and base S_k (see Figure 5). Applying (2.7) and (2.8) yields

$$|(B_{1/2+2^{-k}} \setminus B_{1/2+2^{-k-1}}) \cap E| \leq |\mathcal{C}_{k-1} \cap E| \leq |2S_{k-1}|^{\frac{n}{n-1}},$$

and

$$|(B_{1/2+2^{-k}} \setminus B_{1/2+2^{-k-1}}) \cap E| \geq |S_k| \cdot (2^{-k} - 2^{-k-1}) = 2^{-k-1} |S_k|.$$

Therefore,

$$|S_k| \leq 2^{k+1+\frac{n}{n-1}} |S_{k-1}|^{1+\epsilon}, \quad \epsilon = \frac{1}{n-1}.$$

This yields a contradiction for δ_0 sufficiently small, completing the proof.

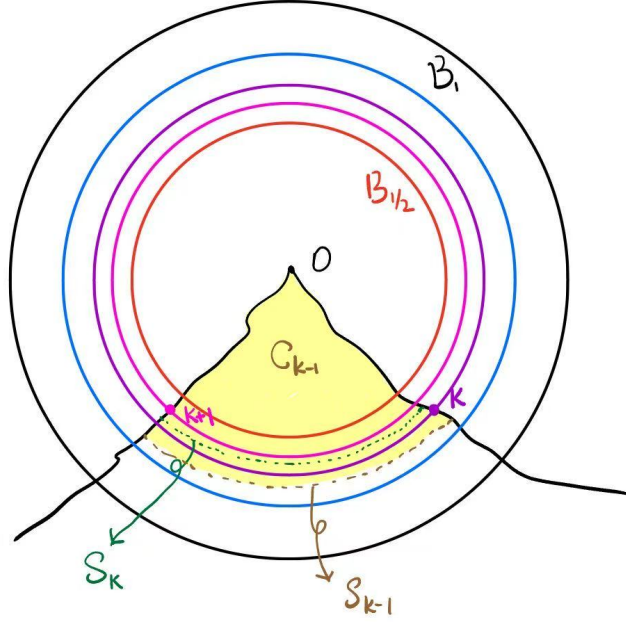


FIGURE 5. Cut-off surfaces.

□

Note that the theorem and the iteration process are analogous to De Giorgi's first step, i.e., Lemma 1. In fact, one may interpret the L^2 norm of a function v on a domain E as a volume measure,

$$\int_E (v)^2 \sim |E|,$$

while the energy $\|\nabla v\|_{L^2}$ corresponds to a perimeter,

$$\int_E |\nabla v|^2 \sim P(E).$$

Thus, Lemma 1 can be viewed as a density estimate in the functional setting:

$$\|v\|_{L^2} \leq \delta_0 \ll 1 \iff |B_1 \cap E| \leq \delta_0 \ll 1 \implies \|v\|_{L^\infty(B_{1/2})} \leq 1 \iff B_{1/2} \cap E = \emptyset.$$

This observation similarly applies to general *stationary* varifolds, which are generalized minimal surfaces. For further extensions in geometric measure theory, particularly the Allard regularity theory, we refer the interested reader to [All72] or [FX03].

2.5. Moser's alternative approach. At the end of this section, we briefly outline Moser's alternative approach [Mos60] to the regularity theory. For a more detailed comparison between Moser's and De Giorgi's methods, see [HL11].

Moser's argument consists of two main parts. The first part achieves the same goal as De Giorgi's, namely lifting integrability to L^∞ , but it replaces the iteration of energy and Sobolev inequalities by a more refined *integration-by-parts* technique. It should be noted, however, that the divergence structure of the equation remains essential here; the method does not extend to non-divergence type equations. The second part is Moser's *weak* Harnack inequality, which later inspired the Krylov–Safonov theory (compare with Corollary 3). Again, this part relies crucially on the divergence form and cannot be directly transferred to the non-divergence setting.

PART I.

Lemma 3 (Moser, 1960 [Mos60]). Let $v \geq 0$ be a *subsolution* of (2.4) in B_1 , i.e.,

$$a_{ij}(x)D_{ij}v \geq 0.$$

Then for every $p > 0$,

$$\sup_{B_{1/2}} v \leq C \|v\|_{L^p(B_1)},$$

where C depends on p, n, λ, Λ .

Proof. We divide the proof into four steps.

Step 1 (The case $p \geq 2$) For a nonnegative test function $\varphi \in H_0^1(B_1)$ we have

$$\int a_{ij} D_i v D_j \varphi \geq 0.$$

Choose $\varphi = \eta^2 v^{p-1}$ with $\eta \in C_c^\infty(B_1)$, $\eta \geq 0$. Inserting it into the inequality gives

$$(p-1) \int a_{ij} D_i v D_j v \cdot v^{p-2} \eta^2 \leq -2 \int a_{ij} D_i v D_j \eta \cdot v^{p-1} \eta.$$

Using the uniform ellipticity, we obtain

$$\int |\nabla v|^2 v^{p-2} \eta^2 \leq C \int |\nabla v| |\nabla \eta| \eta v^{p-1}.$$

Rewrite the right-hand side as $\int \left(|\nabla v| v^{\frac{p-2}{2}} \eta \right) \left(|\nabla \eta| v^{\frac{p}{2}} \right)$, and apply Young inequality $ab \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2$, we obtain

$$\int |\nabla v|^2 v^{p-2} \eta^2 \leq C_\varepsilon \int |\nabla \eta|^2 v^p.$$

Now observe that $|\nabla(v^{p/2})|^2 = \frac{p^2}{4} v^{p-2} |\nabla v|^2$; hence

$$(2.9) \quad \int |\nabla(v^{p/2})|^2 \eta^2 \leq C_1 \int |\nabla \eta|^2 v^p,$$

where C_1 is up to $\varepsilon, p, \lambda, \Lambda$.

Step 2 (Sobolev embedding) Applying the Sobolev inequality to the function $\eta v^{p/2}$ gives

$$\left(\int (\eta v^{p/2})^{\frac{2n}{n-2}} \right)^{\frac{n-2}{n}} \leq C_S \int (|\nabla \eta| v^{p/2} + \eta |\nabla(v^{p/2})|)^2.$$

Combining with (2.9) and using Young inequality again, we obtain

$$(2.10) \quad \left(\int v^{\frac{n}{n-2} \cdot p} \cdot \eta^{\frac{2n}{n-2}} \right)^{\frac{n-2}{n}} \leq C_2 \int |\nabla \eta|^2 v^p,$$

where C_2 depends on p, n, λ, Λ for sufficiently small ε .

Step 3 (Iteration) We define cut-off functions $\eta_k \in C_c^\infty(B_1)$ such that

$$\eta_k := \begin{cases} 1, & \text{in } B_{1/2+2^{-k}}, \\ 0, & \text{outside } B_{1/2+2^{-k+1}} \end{cases}, \quad k = 1, 2, \dots$$

Set $p_k = p \left(\frac{n}{n-2} \right)^k$ (so that $p_k \rightarrow \infty$). Applying (2.10) with $\eta = \eta_k$ and exponent p_k yields

$$\int_{B_{1/2+2^{-k}}} v^{p_k} \lesssim C 2^{2k(1+\epsilon)} \left(\int_{B_{1/2+2^{-k+1}}} v^{p_{k-1}} \right)^{1+\epsilon}, \quad \epsilon = \frac{2}{n-2} > 0,$$

which leads to the same argument in the proof of Lemma 1. Thus, we obtain

$$(2.11) \quad \sup_{B_{1/2}} v \leq C_0 \|v\|_{L^p(B_1)} \quad p \geq 2.$$

Step 4 (The case $0 < p < 2$) By (2.11) with $p = 2$ and Young inequality, we obtain

$$\sup_{B_{1/2}} v \lesssim \epsilon \sup_{B_1} v^{1-\frac{p}{2}} + C_\epsilon \left(\int v^p \right)^{1/2}.$$

Finally, by absorbing lemma, we complete the proof. □

PART II.

Lemma 4 (Moser weak Harnack inequality, 1960 [Mos60]). Let $v \geq 0$ be a *supersolution* of (2.4) in B_1 , i.e.,

$$a_{ij}(x) D_{ij} v \leq 0.$$

Then there exist constants $p_0 > 0$ and $C > 0$, depending only on n, λ, Λ , such that

$$\inf_{B_{1/2}} v \geq C \|v\|_{L^{p_0}(B_1)}.$$

Sketch of proof. Assume without loss of generality that $v > 0$ in B_1 . Apply Lemma 3 to v^{-1} , which is nonnegative and satisfies $a_{ij}D_{ij}(v^{-1}) \geq 0$ (*subsolution*). For any $p > 0$,

$$(2.12) \quad \inf_{B_{1/2}} v \gtrsim \left(\int v^{-p} \right)^{-1/p}.$$

Note that the function $w := -\log v$ is a *subsolution*. By the John-Nirenberg inequality [JN61], there exists $p_0 = p_0(n, \lambda, \Lambda) > 0$ such that

$$\int e^{p_0|w-\bar{w}|} \leq C,$$

where $\bar{w} = \frac{1}{|B_1|} \int w$. Take $p = p_0$ in (2.12). $\square$

2.6. Notes. The regularity theory discussed so far relies on the uniform ellipticity condition (2.1). A natural question is whether this condition can be relaxed to mere *strict* ellipticity (or *strict* convexity of F). Under suitable boundary conditions, minimizers of $\mathcal{E}$ are then Lipschitz, i.e., $\|\nabla u\|_{L^\infty} < \infty$. Whether such minimizers are necessarily C^1 remained open for decades. De Silva and Savin [DSS10] answered this positively in dimension $n = 2$, by assume that the set of degenerate points is at most finite. For $n \geq 4$, Mooney [MS16, Moo20] constructed counterexamples. The case $n = 3$ remains open; see [Moo22] for further discussion.

The ideas behind De Giorgi's iteration have been extended to many other settings. A prominent variant is the " ϵ -regularity" theory: there exists a universal $\epsilon > 0$ such that if the energy of a solution is smaller than ϵ , then the solution is bounded. In geometric measure theory, this philosophy appears in Allard's regularity theorem for minimal surfaces, where the relevant energy—called the *excess*—measures the deviation of an approximate tangent plane of the rectifiable set from a fixed direction. On the other hand, Moser's approach was applied by Schoen–Simon–Yau [SSY75] (see also Theorem 2.21 of [CM11]) to obtain L^p estimates for minimal surfaces, replacing the L^2 estimates (stability) given by Simons' inequality [Sim68] (Lemma 2.1 of [CM11]). This led to a

solution of the stable Bernstein problem under a volume growth assumption for $n \leq 5$, with a dimensional gap for the exponent p . Recently, [Bel25] used De Giorgi's iteration to extend the result to $n = 6$.

The method has since been adapted to various degenerate and nonlocal settings, such as the p -Laplacian, degenerate parabolic equations, and fractional diffusion, and has found applications in diverse fields including fluid dynamics [CV10] and geometric analysis; see for instance [Vas16, BM25, Vio23] for further discussions.

3. FULLY NONLINEAR EQUATIONS AND KRYLOV-SAFONOV THEORY

3.1. Introduction to fully nonlinear equations. In the next two sections, we study the regularity theory of *fully nonlinear* equations,

$$(3.1) \quad F(D^2u) = 0.$$

Key results were established by Krylov-Safonov [KS79, KS81] for $C^{1,\alpha}$ regularity, and by Evans-Krylov [Eva82, Kry83, Kry84] for $C^{2,\alpha}$ regularity.

Throughout, we assume F is *uniformly elliptic*: there exist constants $\lambda, \Lambda > 0$ such that

$$\lambda\|N^+\| - \Lambda\|N^-\| \leq F(M + N) - F(M) \leq \Lambda\|N^+\| - \lambda\|N^-\|,$$

which locally is equivalent to

$$\lambda\delta_{ij} \leq F_{ij} \leq \Lambda\delta_{ij}, \quad F_{ij} := \frac{\partial F}{\partial u_{ij}}.$$

Assuming $F(0) = 0$ without loss of generality, we have

$$\lambda\|M^+\| - \Lambda\|M^-\| \leq F(M) \leq \Lambda\|M^+\| - \lambda\|M^-\|.$$

Thus, the uniform ellipticity of F implies that for any “solution” u of (3.1), its Hessian satisfies $\|(D^2u)^+\| \sim \|(D^2u)^-\|$. Consequently, there exist pointwise-defined coefficients

$\{a_{ij}(x)\}$ (merely measurable and bounded) such that

$$(3.2) \quad a_{ij}(x)D_{ij}u = 0, \quad \lambda\delta_{ij} \leq a_{ij} \leq \Lambda\delta_{ij},$$

which is equivalent to the implicit form (3.1).

However, unlike the divergence-type equations studied earlier, which admit a natural weak formulation via integration by parts.³ Crandall–Lions [CL83] and Evans [Eva78, Eva80] introduced the concept of viscosity solutions for fully nonlinear equations, which serves as a natural replacement for the weak solution notion used in divergence-type equations. We say that u is a *viscosity supersolution* to (3.2) in a domain Ω if for any test function $\varphi \in C^2(\Omega)$ and any point $x_0 \in \Omega$ where φ touches u from below (i.e., $\varphi(x_0) = u(x_0)$ and $\varphi(x) \leq u(x)$ for all x in a neighborhood of x_0), we have

$$a_{ij}(x_0)D_{ij}\varphi(x_0) \leq 0.$$

Analogously, u is a *viscosity subsolution* if for any $\varphi \in C^2(\Omega)$ touching u from above at x_0 (i.e., $\varphi(x_0) = u(x_0)$ and $\varphi(x) \geq u(x)$ near x_0), we have

$$a_{ij}(x_0)D_{ij}\varphi(x_0) \geq 0.$$

Finally, u is a *viscosity solution* if it is both a viscosity supersolution and a viscosity subsolution. We refer readers to [CC95, FRRO23] for more details.

The question is whether u enjoys the same $C^{1,\alpha}$ regularity as in De Giorgi’s result, and further whether it is smooth. Indeed, differentiating (3.1) in a direction $e \in \mathbb{S}^{n-1}$ yields

$$(3.3) \quad F_{ij}(D^2u)D_{ij}u_e = 0.$$

³Hence they are often called “integral” or “global” equations, the fully nonlinear equation (3.2) is inherently local and does not possess a straightforward weak solution concept.

Krylov-Safonov proved that $u_e \in C^{1,\alpha}$; however, this alone cannot initiate a bootstrap argument because the coefficients F_{ij} depend on second-order derivatives. Evans-Krylov established higher $C^{2,\alpha}$ regularity under the additional assumption that F is concave, which does start the bootstrap and leads to smooth solutions. Both theories will be presented in this paper.

For simplicity, set $F_{ij} = a_{ij}(x)$, where a_{ij} are merely measurable and bounded, and let $v = u_e$. We prove that v , which solves

$$(3.4) \quad a_{ij}(x)D_{ij}v = 0, \quad \lambda\delta_{ij} \leq a_{ij}(x) \leq \Lambda\delta_{ij},$$

belongs to C^α . More generally, we state the Krylov-Safonov *Harnack* inequality (**Krylov-Safonov theory**), whose proof will be divided into two parts.

Theorem 3 (Krylov-Safonov, 1979 [KS79, KS81]). Let v be a nonnegative solution to (3.4) in B_1 , where a_{ij} are *uniformly elliptic*, measurable and bounded. Then

$$\sup_{B_{1/2}} v \leq C \inf_{B_{1/2}} v,$$

where C depends only on n, λ, Λ .

Intuitively, one may view the viscosity solution v as a kind of *harmonic* function.

Observe that the L^ϵ -norm of a superharmonic function is controlled by its value at a point; hence, if we fix the value of v at some point, say $v(0) = 1$ without loss of generality, then v belongs to L^ϵ . See Lemma 4 and Lemma 7 below.

Conversely, the value of a subharmonic function is controlled by the nearby L^ϵ -norm, so a fixed L^ϵ -norm may govern the value at the center. See Lemma 1, Lemma 3 and Lemma 8 below.

Combining these two facts, one can propagate control on v from a larger domain to a smaller one, which yields an estimate on the oscillation. These correspond to the two parts of the proof.

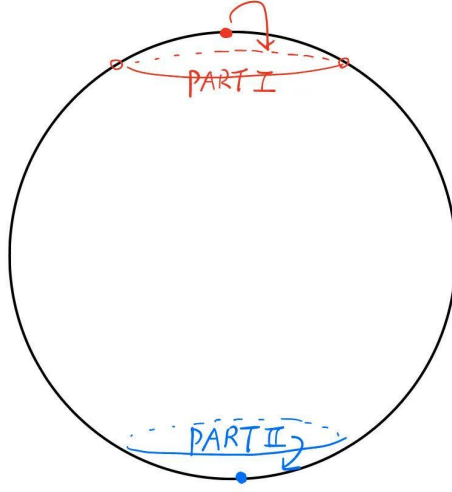


FIGURE 6. mean value inequalities

3.2. Two main tools. Compared with the divergence-type equation (2.4), the equation (3.4) lacks a crucial ingredient: an energy inequality. This absence is not merely a technical restriction⁴. Consequently, the iteration argument used for divergence-type equations fails in the nondivergence case, and one must instead exploit intrinsic structures of elliptic equations, such as the *maximum principle* or comparison principle.

Lemma 5 (Alexandroff–Bakelman–Pucci, 1960s [Ale58, Bak83, PS07]). Let v satisfy

$$a_{ij}(x)D_{ij}v = f \quad \text{in } B_1,$$

with $v \geq 0$ on ∂B_1 . Then

$$\sup(v^-)^n \lesssim \int_{\Gamma^+} |f^+|^n,$$

where Γ^+ denotes the *contact set* of v .

Remark 8. The contact set Γ^+ is defined as

$$\Gamma^+ := \{y \in B_1 : p(y) \cdot (x - y) \leq v(x) - v(y), \text{ for some plane } p(y) \text{ through } y\}.$$

⁴Indeed, the energy $|\nabla u|$ is naturally controlled by the functional $\mathcal{E}$. Also, see Lemma 3.

The primary utility of the ABP estimate lies in obtaining precise bounds for the *measure* of the contact set.

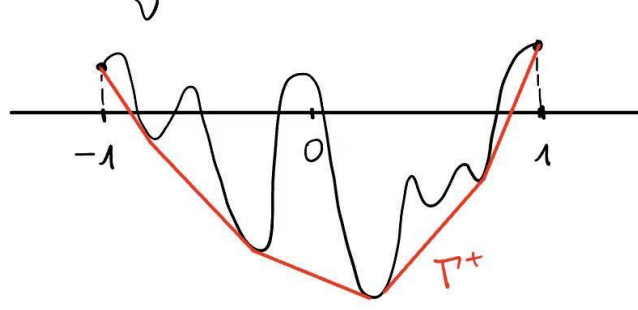


FIGURE 7. Contact set.

A detailed proof can be found in standard PDE textbooks [GT77, HL11]; we provide a brief sketch.

Sketch of proof. First, by a change of variables we have

$$\int_{Dv(\Gamma^+)} g \leq \int_{\Gamma^+} g(Dv) |\det D^2 v|, \quad \forall \text{nonnegative } g \in L^1_{\text{loc}}.$$

Then, by moving planes to contact points, one can show that the ball $B_{\tilde{M}}(0)$ is contained in $Dv(\Gamma^+)$, where $\tilde{M} \sim \sup v^-$. Choosing an appropriate g completes the proof. $\square$

Second, we state a Calderón–Zygmund type lemma.

Lemma 6 (A–B lemma). Let $A \subset B \subset \mathbb{Q}_1$, where $\mathbb{Q}_1$ is the unit cube. Assume

- (i) $|A| \leq \delta$;
- (ii) For any sub-cube $Q \subset \mathbb{Q}_1$, if $\frac{|A \cap Q|}{|Q|} > \delta$, then the dilated cube $\tilde{Q}$ (for example, obtained by expanding each edge by a factor of 3) is contained in B .

Then $|A| \leq \delta |B|$.

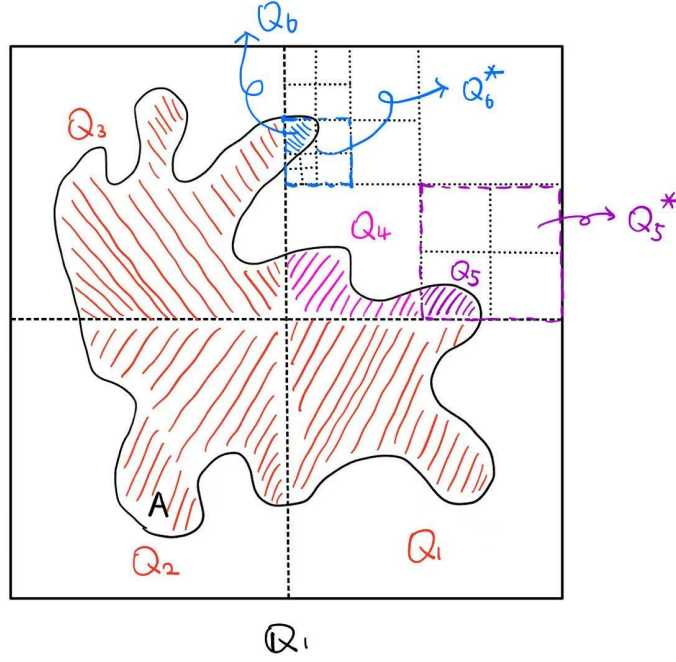


FIGURE 8. Calderón–Zygmund decomposition.

Proof. We perform a Calderón–Zygmund decomposition [MS13] of A inside Q_1 (see Figure 8). Let $\{Q_j\}$ be the collection of non-overlapping dyadic sub-cubes such that

$$\frac{|A \cap Q_j|}{|Q_j|} > \delta,$$

and let Q_j^* denote their predecessors (the last cubes in the decomposition that were not selected). By construction,

$$\frac{|A \cap Q_j^*|}{|Q_j^*|} \leq \delta.$$

Condition (ii) implies $\tilde{Q}_j \subset B$, hence $Q_j^* \subset B$. After removing repetitions and reindexing, we obtain a family of disjoint cubes $\{Q_{j'}^*\}$ with $\bigcup Q_{j'}^* \subset B$. Consequently,

$$|A| \leq \left| \bigcup Q_j \right| \leq \left| \bigcup (Q_{j'}^* \cap A) \right| \leq \sum |Q_{j'}^* \cap A| \leq \delta \sum |Q_{j'}^*| \leq \delta |B|,$$

where the last inequality uses the disjointness of the $Q_{j'}^*$. $\square$

3.3. Krylov-Safonov weak Harnack inequality. We now present the first part of the proof of Theorem 3.

Lemma 7 (Krylov-Safonov weak Harnack inequality). Let v be a supersolution to (3.4) with $v(0) = 1$. Then $v \in L_{weak}^\epsilon(B_{1/2})$, i.e.,

$$|\{v > \lambda\} \cap B_{1/2}| \leq \lambda^{-\epsilon}, \quad \forall \lambda > 0.$$

Remark 9. Recall *Chebyshev* inequality.

The proof proceeds in two steps.

First, we derive a contact estimate via the ABP method. To present a baby version, We claim that

$$|\{v < 2\} \cap B_1| \geq \theta > 0,$$

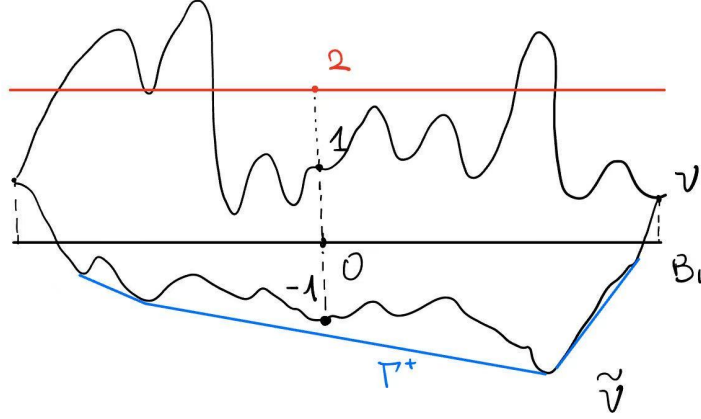
for some universal $\theta > 0$. Define

$$\tilde{v} := v + 2(|x|^2 - 1).$$

Then $\tilde{v}(0) = -1$ and $\tilde{v}|_{\partial B_1} \geq 0$. A direct calculation gives

$$a_{ij}D_{ij}\tilde{v} \leq C,$$

for some constant $C > 0$.



Applying the ABP estimate to $\tilde{v}$ yields

$$1 \leq \sup(\tilde{v}^-)^n \leq \int_{\Gamma^+} C^n.$$

If $x_0 \in \Gamma^+$, then $\tilde{v}(x_0) < 0$ (for more preciser auxiliary functions, seeing the sublemma below), which implies $v(x_0) < 2$. Hence $\Gamma^+ \subset \{v < 2\}$, and therefore

$$1 \leq C^n |\{v < 2\}|.$$

We now state the general version of the measure estimate.

sublemma 4 (Contact estimate). Let $x_0 \in B_{1/4}$ and $M > 0$ be a large constant. Then

$$|\{v < M\} \cap B_{1/8}(x_0)| \geq \theta |B_{1/8}(x_0)|,$$

where $\theta > 0$ is universal.

Proof. Construct an auxiliary function

$$\gamma(x) := \begin{cases} \frac{\epsilon}{2} |x - x_0|^2 - A, & \text{in } B_{1/8}(x_0), \\ -|x - x_0|^{-\sigma} - B, & \text{outside } B_{1/8}(x_0), \end{cases}$$

with constants A, B chosen so that

$$a_{ij}D_{ij}\gamma \leq \begin{cases} C, & \text{in } B_{1/8}(x_0), \\ 0, & \text{outside } B_{1/8}(x_0), \end{cases}$$

and set $\tilde{v} := v + \gamma$ (see Figure 9). We can arrange γ so that $\tilde{v}(0) = -2$ and $\inf \gamma \geq -M$, without loss of generality, set $M > 2$.

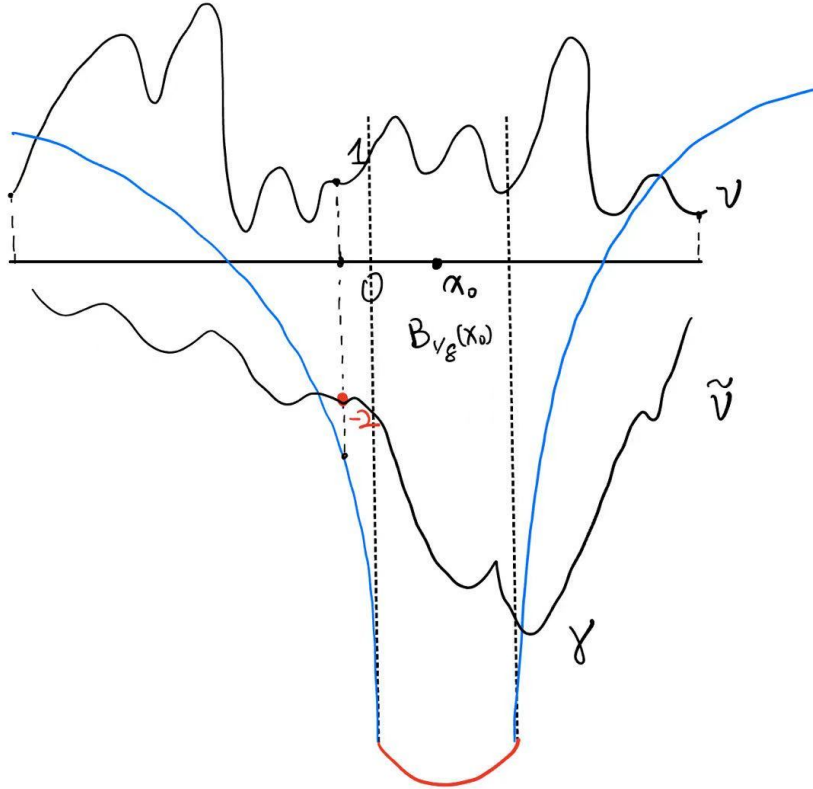


FIGURE 9. Measure estimate.

Consequently,

$$a_{ij}D_{ij}\tilde{v} \leq \begin{cases} C, & \text{in } B_{1/8}(x_0), \\ 0, & \text{outside } B_{1/8}(x_0). \end{cases}$$

Applying the ABP estimate to $\tilde{v}$ yields

$$2^n \leq \sup(\tilde{v}^-)^n \leq C^m |\Gamma^+|.$$

If $x \in \Gamma^+$, then $\tilde{v}(x) < 0$ (it is clear in $B_{1/8}(x_0)$; and Γ^+ can not escape from $B_{1/8}(x_0)$ because $\tilde{v}$ is a supersolution outside $B_{1/8}(x_0)$), which implies $v(x) < M$. Hence $\Gamma^+ \subset \{v < M\} \cap B_{1/8}(x_0)$, and the desired inequality follows. $\square$

The second step provides the weak L^ϵ estimate.

Proof of Lemma 7. Define

$$A_k := \{v > M^k\} \cap B_{1/2}, \quad k = 0, 1, 2, \dots$$

We will show that the pair (A_k, A_{k-1}) satisfies the conditions of the A-B lemma with $\delta = 1 - \theta$.

Assume that for some cube $Q \subset B_{1/2}$ we have

$$(3.5) \quad \frac{|A_k \cap Q|}{|Q|} > \delta,$$

which is equivalent to

$$|\{v \leq M^k\} \cap Q| < \theta |Q|.$$

Let $\tilde{v} = v/M^{k-1}$ and rescale Q to a cube $\mathbb{Q}_{1/8}(x_0)$. Then

$$|\{\tilde{v} \leq M\} \cap \mathbb{Q}_{1/8}(x_0)| < \theta |\mathbb{Q}_{1/8}(x_0)|.$$

We claim that $\tilde{v} > 1$ on the dilated cube $\tilde{\mathbb{Q}}_{1/8}(x_0)$. If not, there exists $\tilde{x} \in \tilde{\mathbb{Q}}_{1/8}(x_0)$ with $\tilde{v}(\tilde{x}) \leq 1$. By Lemma 4 applied to $\tilde{v}$ (after a translation), we obtain

$$|\{\tilde{v} \leq M\} \cap \mathbb{Q}_{1/8}(x_0)| \geq \theta |\mathbb{Q}_{1/8}(x_0)|,$$

contradicting the previous inequality. Hence $\tilde{v} > 1$ on $\tilde{\mathbb{Q}}_{1/8}(x_0)$, which means $v > M^{k-1}$ there; thus $\tilde{Q} \subset A_{k-1}$.

Therefore, the hypotheses of the A-B lemma are satisfied, and we obtain

$$|A_k| \leq (1 - \theta)|A_{k-1}|.$$

Iterating gives $|A_k| \leq (1 - \theta)^k |A_0|$. Setting $\epsilon = -\log_M(1 - \theta) > 0$, we have

$$|\{v > M^k\}| \leq M^{-\epsilon k},$$

which implies $v \in L_{\text{weak}}^\epsilon(B_{1/2})$. □

Remark 10. One can explicitly estimate the " L^ϵ -norm" in $B_{1/2}$:

$$\begin{aligned} \int_{B_{1/2}} v^\epsilon &\leq M^\epsilon |\{v < M\}| + M^{2\epsilon} |\{M \leq v < M^2\}| + \dots \\ &\leq M^\epsilon + M^{2\epsilon}(1 - \theta) + \dots + M^{k\epsilon}(1 - \theta)^{k-1} + \dots \\ &\leq \frac{M^\epsilon}{1 - M^\epsilon(1 - \theta)} < \infty. \end{aligned}$$

Corollary 3 (L^ϵ -estimate). If $v \geq 0$ is a supersolution of (3.4) in $\mathbb{Q}_3$, then

$$\|v\|_{L^\epsilon(\mathbb{Q}_1)} \lesssim \inf_{\mathbb{Q}_2} v.$$

Remark 11. Compare it with Lemma 4.

The above argument already yields C^α regularity. The proof of the next result follows the notes of Professor Yu Yuan.

Corollary 4 (C^α -estimate). Let v be a nonnegative *solution* of (3.4). Then

$$\|v\|_{C^\alpha(B_{1/2})} \lesssim \|v\|_{L^\infty(B_1)}.$$

Proof. It suffices to show

$$\text{osc}_{\mathbb{Q}_1} v \leq \sigma \text{osc}_{\mathbb{Q}_2} v, \quad \sigma < 1.$$

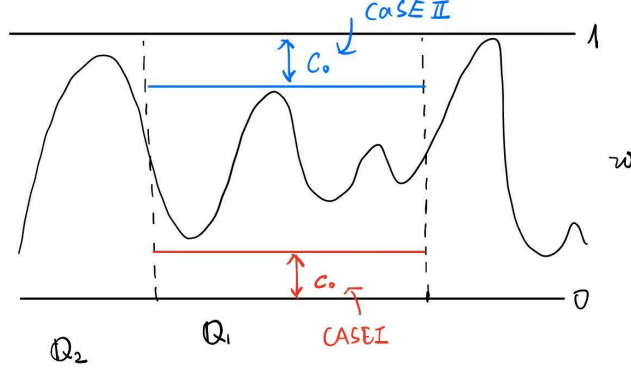


FIGURE 10. Oscillation decay

Define

$$w := \frac{v - \inf_{Q_2} v}{\text{osc}_{Q_2} v}, \quad 0 \leq w \leq 1,$$

which also solves (3.4). We consider two cases; if the first does not hold, we replace w by $1 - w$. Recall that w is a *solution*.

CASE I: $|\{w \geq 1/2\} \cap Q_1| \geq \frac{1}{2}$.

CASE II: $|\{w \geq 1/2\} \cap Q_1| < \frac{1}{2}$.

Assume CASE I holds. Then

$$\frac{1}{2} \left(\frac{|Q_1|}{2} \right)^{1/\epsilon} \leq \left(\int_{Q_1} w^\epsilon \right)^{1/\epsilon} \lesssim \inf_{Q_2} w \leq \inf_{Q_1} w.$$

Hence $\inf_{Q_1} w \geq c_0 > 0$.

□

Remark 12. Note that the proof fails if v is merely a *subsolution* or *supersolution*; this constitutes the main difficulty in the Evans-Krylov theory. Also see Lemma 4.

Remark 13. The proof above naturally leads to a Liouville-type theorem analogous to Corollary 2.

3.4. Local maximum principle. Finally, we present the second part of the proof, which implies Theorem 3.

Lemma 8 (Local maximum principle). Let v be a *subsolution* belonging to $L^\epsilon(B_1)$ with $\|v\|_{L^\epsilon(B_1)} \leq 1$. Then there exists a large constant $M > 0$ such that

$$\sup_{B_{1/2}} v \leq M.$$

Remark 14. Compare it with Lemma 1 and Lemma 3.

Proof. The proof relies on so-called *blow-up* argument. Assume, for contradiction, that there exists $x_0 \in B_{1/2}$ with $v(x_0) = M_0$. We claim that there is a small cube $Q(x_0) \subset B_1$ such that

$$\sup_{Q(x_0)} v \geq (1 + \delta)M_0, \quad \text{for some universal } \delta.$$

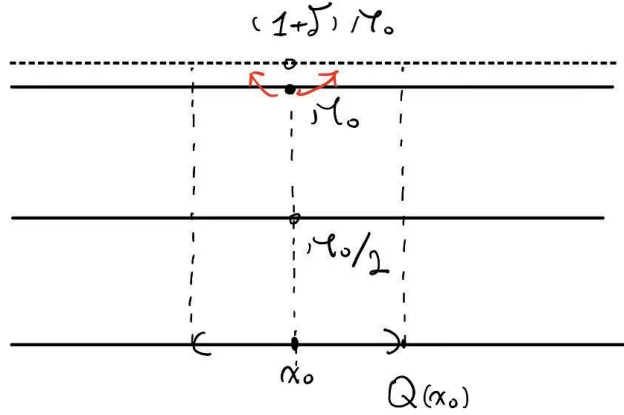


FIGURE 11. blow-up

Since $v \in L^\epsilon(B_1)$ and $\|v\|_{L^\epsilon} \leq 1$,

$$\left| \left\{ v > \frac{M_0}{2} \right\} \cap Q_1 \right| \leq \left(\frac{M_0}{2} \right)^{-\epsilon}.$$

Choose $Q(x_0)$ with $|Q(x_0)| > 2\left(\frac{M_0}{2}\right)^{-\epsilon}$ (recall M may be taken sufficiently large); then

$$\left| \left\{ v > \frac{M_0}{2} \right\} \cap Q(x_0) \right| \leq \frac{1}{2}|Q(x_0)|.$$

Now prove the claim. If not, i.e., $v < (1 + \delta)M_0$ in $Q(x_0)$. Define

$$w := \frac{(1 + \delta)M_0 - v}{\delta M_0},$$

which is a nonnegative supersolution with $w(x_0) = 1$. By K-S weak Harnack inequality (Lemma 7), $w \in L_{weak}^\epsilon(Q(x_0))$. Note that, if $v \leq M_0/2$, then

$$w \geq \frac{(1 + \delta)M_0 - M_0/2}{\delta M_0} = \frac{\frac{1}{2} + \delta}{\delta}.$$

Hence

$$\begin{aligned} \left| \left\{ v \leq \frac{M_0}{2} \right\} \cap Q(x_0) \right| &\leq \left| \left\{ w \geq \frac{\frac{1}{2} + \delta}{\delta} \right\} \cap Q(x_0) \right| \\ &\leq \left(\frac{\delta}{\frac{1}{2} + \delta} \right)^\epsilon |Q(x_0)| < \frac{1}{2} |Q(x_0)|, \end{aligned}$$

for sufficiently small δ . This contradiction establishes the claim.

Hence, one may choose a $x_1 \in Q(x_0)$ such that

$$v(x_1) \geq (1 + \delta)M_0, \quad |x_1 - x_0| \lesssim M_0^{-\epsilon};$$

$$v(x_2) \geq (1 + \delta)^2 M_0, \quad |x_2 - x_1| \lesssim [(1 + \delta)M_0]^{-\epsilon};$$

$$v(x_k) \geq (1 + \delta)^k M_0, \quad |x_k - x_{k-1}| \lesssim [(1 + \delta)^k M_0]^{-\epsilon}, \quad k = 3, 4, \dots$$

Note that

$$\begin{aligned} |x_k - x_0| &\leq |x_k - x_{k-1}| + \dots + |x_1 - x_0| \\ &\leq [1 + (1 + \delta) + \dots + (1 + \delta)^k]^{-\epsilon} M_0^{-\epsilon} \\ &\leq \left(\frac{\delta}{(1 + \delta)^k - 1} \right)^\epsilon M_0^{-\epsilon} < 1/8, \end{aligned}$$

as δ sufficiently small; and $v(x_k) \rightarrow \infty$ as $k \rightarrow \infty$, which leads to the contradiction. $\square$

Remark 15. This yields a contradiction because, starting from a point x_k with large value, one can iteratively produce points arbitrarily close to x_0 with ever larger values, contradicting the definition of L_{weak}^ϵ .

4. CONCAVE PROPERTY AND EVANS-KRYLOV THEORY

4.1. Concave assumption of F and Gap. We have established the Krylov-Safonov theory, i.e., the $C^{1,\alpha}$ estimate for

$$F_{ij}(D^2u)D_{ij}u_e = 0.$$

To obtain $C^{2,\alpha}$ estimates, an additional assumption is required, namely that F is *concave*. Differentiating the equation again in a direction $f \in \mathbb{S}^{n-1}$ yields, formally,

$$F_{ij,kl}(D^2u)D_{ij}u_e D_{kl}u_f + F_{ij}(D^2u)D_{ij}u_{ef} = 0.$$

Since F is concave, we have

$$(4.1) \quad F_{ij}(D^2u)D_{ij}v \leq 0, \quad v := u_{ef}.$$

Thus, v is a *subsolution*. Recall the L^ϵ estimate in Krylov-Safonov theory (Corollary 3); however, there is a gap because we lack a corresponding supersolution. The fact that $u_{ef} \in L^\epsilon$ was proved separately by Evans [Eva85] and Lin [Lin86]. Then, by the local maximum principle (Lemma 8), we obtain $u \in C^{1,1}$. For further details, we refer the reader to [CC95].

It remains to prove that $u \in C^{2,\alpha}$, i.e., $D^2u \in C^\alpha$. The C^α estimate (Corollary 4) requires a "solution", but we only know that v is a subsolution. This gap constitutes the main difficulty in the Evans-Krylov theory. Fortunately, however, the relation

$$(D^2u)^+ \sim (D^2u)^- \iff v^+ \sim v^-$$

suggests that v is almost a *supersolution*, an idea that will be elaborated in the following arguments.

Firstly, we state the main result.

Theorem 4 (Evans-Krylov, 1982 [Eva82, Kry83, Kry84]). Let u be a $C^{1,1}$ solution to

$$F(D^2u) = 0, \quad \text{in } B_1,$$

where F is uniformly elliptic and *concave* (or convex). Then $u \in C^{2,\alpha}(B_{1/2})$, precisely

$$\|u\|_{C^{2,\alpha}(B_{1/2})} \lesssim \|u\|_{L^\infty(B_1)}.$$

Set $\mathcal{H} := D^2u : B_1 \rightarrow \mathcal{S}$ as the Hessian map, where $\mathcal{S}$ denotes the space of real symmetric matrices with the usual Euclidean norm. The theorem follows from the following oscillation lemma. Without loss of generality, assume $\text{diam } \mathcal{H}(B_1) = 1$.

Lemma 9 (Oscillation decay). There exists a $\rho_0 < 1$ such that

$$\text{diam } \mathcal{H}(B_{\rho_0}) < \frac{1}{2}.$$

4.2. Deletion argument. The idea relies on a *deletion* argument: one can remove a small ball from the covering of $\mathcal{H}(B_1)$ without affecting the covering of $\mathcal{H}(B_{1/2})$. This is made precise by the covering lemma stated below.

Let $\mathbf{t} := \sup_{B_1} \lambda_{\max}(D^2u)$.

sublemma 5 (Covering). There exists a universal $0 < \delta \ll 1$ such that $\mathcal{H}(B_1)$ can be covered by finitely many balls $\{B_\delta(M_i)\}$. Moreover, for some direction $g \in \mathbb{S}^{n-1}$ and some ball $B_\delta(M_3)$ we have

$$\mathbf{t} - u_{gg}(x) \gtrsim \frac{1}{8}, \quad \forall x \in \mathcal{H}^{-1}(B_\delta(M_3)),$$

and

$$|\mathcal{H}^{-1}(B_\delta(M_3))| \gtrsim \delta^{n^2}.$$

Proof. Cover $\mathcal{H}(B_1)$ by balls of radius δ , which will be determined at the end of proof; their number satisfies $\# \lesssim \delta^{-n^2}$. Hence some ball $B_\delta(M_3)$ satisfies the measure estimate

$$(4.2) \quad |\mathcal{H}^{-1}(B_\delta(M_3))| \gtrsim \delta^{n^2}.$$

Since $\text{diam } \mathcal{H}(B_1) = 1$, there exists $M \in \mathcal{H}(B_1)$ with $\|M - M_3\| \geq \frac{1}{4}$. Both $F(M) = F(M_3) = 0$. By uniform ellipticity,

$$\lambda\|(M - M_3)^+\| - \Lambda\|(M - M_3)^-\| \leq 0 \leq \Lambda\|(M - M_3)^+\| - \lambda\|(M - M_3)^-\|,$$

which implies $\|(M - M_3)^+\| \sim \|(M - M_3)^-\|$. Consequently,

$$\|(M - M_3)^+\| \gtrsim \frac{1}{8}, \quad \|(M - M_3)^-\| \gtrsim \frac{1}{8}.$$

Take points $x_3, y \in B_1$ with $D^2u(x_3) = M_3$ and $D^2u(y) = M$. Choose a unit vector g (e.g., an eigenvector of $(M - M_3)^+$ corresponding to its largest eigenvalue) such that

$$u_{gg}(y) - u_{gg}(x_3) \gtrsim \frac{1}{8}.$$

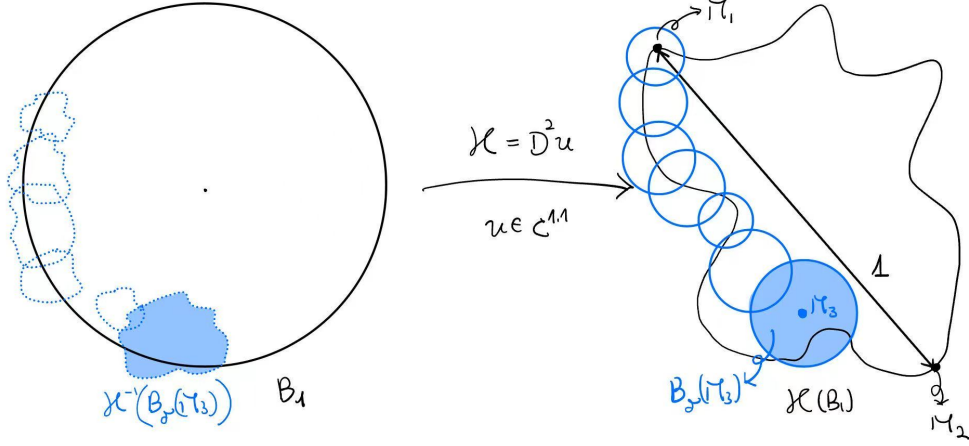
Since $\mathbf{t} \geq u_{gg}(y)$,

$$\mathbf{t} \gtrsim u_{gg}(x_3) + \frac{1}{8}.$$

For any $z \in \mathcal{H}^{-1}(B_\delta(M_3))$, we have $|u_{gg}(z) - u_{gg}(x_3)| \lesssim \delta$. Choosing δ sufficiently small (universally) gives

$$\mathbf{t} - u_{gg}(z) \gtrsim \frac{1}{8} - \delta \gtrsim \frac{1}{8},$$

while preserving (4.2). □



Now consider the function

$$\tilde{v} := \mathfrak{t} - u_{gg},$$

which satisfies:

- (i) $\tilde{v} \geq 0$ in B_1 ;
- (ii) $\tilde{v}$ is a *supersolution*;
- (iii) $\tilde{v} \gtrsim \frac{1}{8}$ on $\mathcal{H}^{-1}(B_\delta(M_3))$.

By (i), (ii) and the L^ϵ estimate (Corollary 3),

$$\inf_{B_{1/2}} \tilde{v} \gtrsim \left(\int_{B_1} \tilde{v}^\epsilon \right)^{1/\epsilon} \gtrsim \frac{1}{8} \delta^{n^2/\epsilon} =: \theta,$$

where the last bound uses (iii). Hence

$$u_{gg}(x) \leq \mathfrak{t} - \theta \quad \forall x \in B_{1/2}.$$

We now prove Lemma 9 via a *deletion* argument.

Proof of Lemma 9. First, cover $\mathcal{H}(B_1)$ by finitely many balls of radius θ (which is smaller than the earlier δ). The estimate above shows that $u_{gg}(x) \leq \mathfrak{t} - \theta$ on $B_{1/2}$; therefore $\mathcal{H}(B_{1/2})$ can be covered by the same family *without* the ball $B_\theta(\mathfrak{t})$ (see Figure 12). Thus one ball is effectively deleted from the covering.

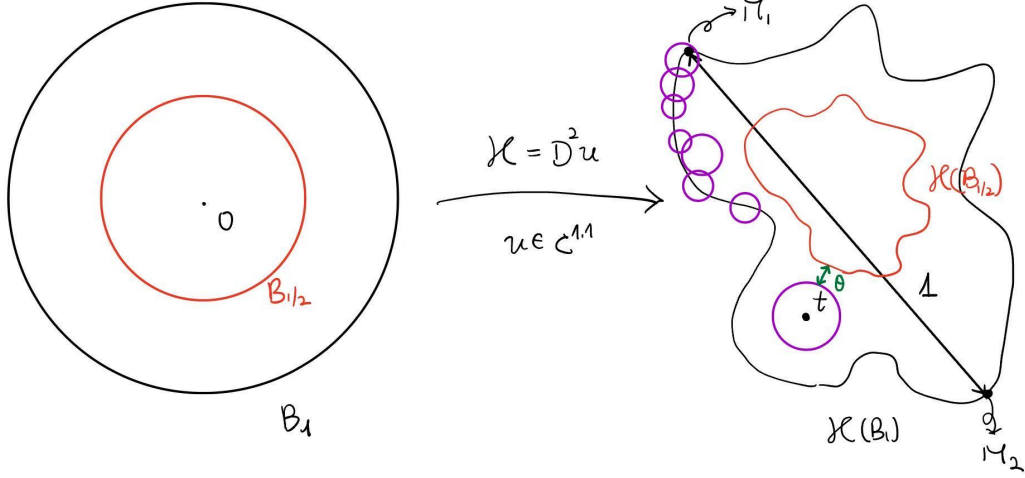


FIGURE 12. deletion

If after this deletion we have $\text{diam } \mathcal{H}(B_{1/2}) \leq \frac{1}{2}$, we may take $\rho_0 = 1/2$ and the lemma is proved. Otherwise, $\text{diam } \mathcal{H}(B_{1/2}) > \frac{1}{2}$, and we can apply Sublemma 5 to the rescaled function $v_1(x) := 2^2 v(x/2)$. Since the number of balls in the covering is finite, after iterating this procedure a finite number k_0 of times we obtain

$$\text{diam } \mathcal{H}(B_{2^{-k_0}}) < \frac{1}{2}.$$

Taking $\rho_0 = 2^{-k_0}$ completes the proof. $\square$

4.3. Notes.

4.3.1. *Krylov-Safonov.* A simpler proof of the Krylov-Safonov theorem can be found in [Moo19]. The ideas behind their Harnack inequality were further developed by Caffarelli [Caf87] and Caffarelli-Wang [CW93], and later extended by Savin [Sav07] to nearly flat solutions (small *perturbation*).

4.3.2. *Evans-Krylov.* Since the Evans-Krylov theory assumes concavity of F , a natural question is whether solutions are always C^2 . The answer is affirmative for $n = 2$ [Nir53]

(see also Theorem 4.8 of [FRRO23]), where solutions are even $C^{2,\alpha}$. Counterexamples for $n \geq 5$ were constructed by Nadirashvili-Vladuts [NV07, NV08, NV13]. The cases $n = 3, 4$ remain open. We refer to [CC95] for a comprehensive treatment of fully nonlinear equations.

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